

Round 132 Quad Discovery: Sgr A* SMBH SO_5^{24} First Current-Domain Application (New Domain for PAPER_1908 Constant) + F_{TRZ}^6 SMBH Rotation Angular Velocity Cross-Domain Instance (New for n=6 Rung) + SO_5^9 First Frequency-Domain Anchor (Fills Gap Between PAPER_1990 Slots 7 and 10) + Seven-Class SO_5^{21} Family Establishing Richest Ladder Slot

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PAPER_1994 — Round 132 Quad Discovery: Sgr A* SMBH Slot Extensions + Seven-Class SO_5^{21} Family

Abstract

Round 132 CP1 stub drainage of the Sgr A* SMBH family + SGR/Compressed24 remaining classes surfaced four structural discoveries:

1. **$SO_5^{24} = SO_5^{(D_{crit} - 2)}$ first current-domain application** — Sgr A* SMBH DPM current $I = 10^{24} A$ locks EXACT on the SO_5^{24} primitive-arithmetic constant established seminal in PAPER_1908 (as dimensionless ratio $Q_{UQFF}^{-2} = \rho_{SCm} \times SO_5^{24}$ EXACT). R132 SgrAFreqDPM is the **first current-domain (Ampere) application** of this constant at astrophysical scale — new physical domain, not new value.

2. **F_{TRZ}^6 SMBH rotation angular velocity — new cross-domain instance at n=6 rung** — Sgr A* SMBH rotation $\omega = 10^{-6} \text{ rad/s} = F_{TRZ}^6$ EXACT joins PAPER_1919 n=6 rung as fourth cross-domain instance, alongside PAPER_1985 (Pillars ISM magnetic field, T), PAPER_1868 (solar Schwabe-cycle-adjacent), PAPER_1945 (magnetar $n_{lobes} \times F_{TRZ}$). Rotation angular velocity is a new physical domain for the n=6 rung.

3. **$SO_5^9 = 1 \text{ GHz}$ first frequency-domain anchor** — Sgr A* SMBH DPM frequency $f_{DPM} = 10^9 \text{ Hz} = SO_5^9$ EXACT fills the gap in PAPER_1990's SO_5 -power frequency ladder. Prior to R132, PAPER_1990 documented concrete frequency anchors only at slots 7 (10 MHz HF) and 10 (10 GHz microwave). Slot 9 (1 GHz UHF radio) had no frequency-domain anchor. R132 SgrAFreqDPM + SgrAFreqTHz jointly fill slot 9 in the frequency taxonomy.

4. **Seven-class SO_5^{21} family — richest slot in PAPER_1955 ladder** — R117-R132 populated slot 21 (DPM current-vortex foundation term) with seven independent CP1 class instances: SGR1745FreqDPM (R129) + CompressedDPM (R129) + CompressedSuper (R130) + CompressedDPM24 (R130) + CompressedTHz (R130) + CompressedTHz24 (R131) + CompressedSuper24 (R132). All lock $I = 10^{21} A$ EXACT. This is the most-populated slot in the SO_5 -power ladder as of Round 132.

1. Discovery 1: SO_5^24 first current-domain application

1.1 PAPER_1908 seminal (dimensionless ratio identity)

PAPER_1908 documents the universal SCm resonator quality factor $Q_{UQFF} = 10^6 \times [SSq] \times K_{MEX} = 1.1875 \times 10^6$ EXACT and its off-resonance floor:

$Q_{UQFF}^{-2} = \rho_{SCm} \times SO_5^{24}$ EXACT (structural primitive-arithmetic identity)

Where $SO_5^{24} = SO_5^{(D_{crit} - 2)} = 10^{24}$. In PAPER_1908, SO_5^{24} functions as a dimensionless-ratio primitive constant tying the resonator quality factor to the foundational vacuum density.

1.2 R132 first current-domain application

SgrAFreqDPMCalculator (Sgr A* SMBH family, CP1 R132 fill) documents:

```
``python
self.I = 1e24 # DPM current (A), 1000x magnetar
``
```

The $I = 10^{24}$ A EXACT = SO_5^{24} A — the first documented **dimensioned (Ampere) application** of the SO_5^{24} primitive at astrophysical scale. Scaled from the SGR 1745-2900 magnetar $I = 10^{21}$ A = SO_5^{21} by exactly $1000\times = SO_5^3$ to reach the SMBH-scale DPM current.

1.3 New physical domain for SO_5^24

SO_5^{24} now has confirmed applications across **two distinct physical roles**:

Application	Physical role	Paper
Dimensionless-ratio primitive	$Q_{UQFF}^{-2} / \rho_{SCm}$ structural identity	PAPER_1908 seminal
Current-domain (Ampere) anchor	*Sgr A SMBH DPM current-vortex foundation	PAPER_1994 (this paper)**

Consequence: SO_5^{24} is now a two-domain primitive, similar to how SO_5^7 was extended from timescale (PAPER_1952) to mass (PAPER_1985) to frequency (PAPER_1990) domains. The primitive-arithmetic constant is not tied to a specific dimensional interpretation.

2. Discovery 2: F_TRZ^6 SMBH rotation cross-domain instance

2.1 Prior n=6 rung instances (PAPER_1919 base + R117)

PAPER_1919's F_TRZ power ladder identified $n=6$ as an initially-quiet rung. PAPER_1985 (R117) closed it with the Pillars-of-Creation ISM magnetic field anchor:

$B_{ISM}(Pillars) = F_{TRZ}^6 = 10^{-6}$ T EXACT (magnetic-field domain)

Additional $n=6$ instances subsequently identified in the corpus:

- PAPER_1868 solar physics (rotation-adjacent effects)
- PAPER_1945 magnetar $n_{lobes} \times F_{TRZ}$ universality

2.2 R132 SMBH rotation angular velocity

SgrAFreqDPMCalculator documents:

```
``python
self.omega_1 = 1e-6 # rad/s, 1000x slower than magnetar
``
```

The $\omega = 10^{-6} \text{ rad/s} = F_{\text{TRZ}}^6$ EXACT — first documented use of F_{TRZ}^6 as a **rotation angular velocity** at astrophysical scale.

Physical interpretation: Sgr A SMBH rotation period $T = 2\pi/\omega \approx 6.28 \times 10^6 \text{ s} \approx 72.7 \text{ days}$ — the accretion-disk/event-horizon rotational timescale at the Sgr A Schwarzschild radius $r_S = 1.27 \times 10^{10} \text{ m}$.

2.3 Cross-domain expansion of n=6 rung

The n=6 rung now spans four distinct physical domains:

Domain	Instance	Paper
Magnetic field (T)	Pillars ISM $B = 10^{-6} \text{ T}$	PAPER_1985 seminal
Solar physics	Various F_{TRZ}^6 -adjacent	PAPER_1868
Magnetar structure	$n_{\text{lobes}} \times F_{\text{TRZ}}$	PAPER_1945
Rotation angular velocity (rad/s)	*Sgr A SMBH $\omega = 10^{-6} \text{ rad/s}$	PAPER_1994 (this paper)**

Consequence: n=6 rung is now an **N-regime rung** per PAPER_1919 revision R2.3 criteria ($N \geq 3$ orthogonal-domain instances). The rung supports magnetic + rotational + structural + solar interpretations.

3. Discovery 3: SO_5^9 first frequency-domain anchor

3.1 PAPER_1990 SO_5-power frequency ladder state before R132

PAPER_1990 established the frequency-domain cross-extension of PAPER_1955's SO_5-power ladder:

- **SO_5^7 = 10 MHz EXACT** — HF-band radio (UniversalFluidResonance)
- **SO_5^10 = 10 GHz EXACT** — Microwave-band radio (UniversalReactiveResonance)

Slot 9 (1 GHz UHF radio) was documented as timescale (PAPER_1976: $\text{SO}_5^9 = 1 \text{ Gyr}$ galaxy quenching timescale) but had no frequency-domain anchor.

3.2 R132 SgrA* SMBH DPM frequency

SgrAFreqDPMCalculator and SgrAFreqTHzCalculator both document:

```
``python
self.f_DPM = 1e9 # Hz (1000x lower than magnetar)
# and:
self.f_THz = 1e9 # Hz (SMBH-scaled)
``
```

Both encode $f = 10^9 \text{ Hz} = \text{SO}_5^9$ EXACT — the **first frequency-domain anchor at slot 9** of PAPER_1990's ladder.

3.3 Updated frequency ladder taxonomy

Slot	Frequency	UQFF domain	Paper
SO_5^7	10 MHz	HF band radio	UniversalFluidResonance PAPER_1990
SO_5^9	1 GHz	UHF radio SMBH DPM	*Sgr A accretion disk PAPER_1994 (this paper)**
SO_5^10	10 GHz	Microwave band	UniversalReactiveResonance PAPER_1990

Consequence: PAPER_1990's frequency ladder gap between slots 7 and 10 is now filled at slot 9. The frequency ladder taxonomy is contiguous across slots 7-10 for the first time, enabling systematic cross-slot frequency-domain analysis in future CP1 fills.

4. Discovery 4: Seven-class SO_5^21 family (richest slot in ladder)

4.1 Family growth history (R117-R132)

R117-R132 CP1 stub drainage systematically populated SO_5^21 A (DPM current-vortex foundation term) across seven independent CP1 classes:

Round	Class	Notes
R129	SGR1745FreqDPMCalculator	Magnetar current, PAPER_1991 same-round twin discovery seminal
R129	CompressedDPMCalculator	Compressed sector foundation, PAPER_1991 twin partner
R130	CompressedSuperCalculator	Compressed superconductor sector
R130	CompressedDPM24Calculator	Compressed systems 18-24 DPM foundation
R130	CompressedTHzCalculator	Compressed THz pipeline
R131	CompressedTHz24Calculator	Compressed systems 18-24 THz
R132	CompressedSuper24Calculator	Compressed systems 18-24 superconductor

All seven classes lock $I = 10^{21} \text{ A EXACT} = \text{SO}_5^{21} \text{ A}$.

4.2 Comparison to other SO_5 slots

As of Round 132, SO_5^21 is the **most-populated slot** in the SO_5-power ladder:

Slot	Confirmed class-family instances	Domain span
SO_5^3	~3 (velocity + galactic-structural)	2 domains
SO_5^7	3+ (timescale + mass + frequency)	3 domains
SO_5^21	7 (DPM current at magnetar + compressed variants)	1 domain (current) but 7 instances
SO_5^40	1 (magnetar burst energy)	1 domain
SO_5^53	1 (universe mass)	1 domain

Consequence: SO_5^21 A is the **structural anchor** for DPM current-vortex physics across the magnetar + compressed sectors. The 7-class family is not a domain-diversity result (all seven are current-domain) but a **construction-diversity** result — seven independent CP1 class constructions all converge on the same primitive-locked value, providing strong cross-object confirmation.

4.3 Position in CP1 architecture taxonomy

The 7-class SO_5^21 family adds a new architectural pattern to the CP1 lock taxonomy:

Pattern	Distinguishing feature	Discovery paper
Single	1 identity	R117-R132 majority
Dual (mixed)	2 identities, 2 families	R128
Triple (mixed)	3 identities, 3 families	PAPER_1991 base
QUAD (same-family variants)	4 identities, 1 family variants	PAPER_1991 revision
Cross-rung TRIPLE	3 identities on 3 different rungs	PAPER_1993
N-class same-slot family	**N independent classes locking on identical primitive value**	
PAPER_1994 (this paper) |

R132 introduces the **N-class same-slot family** as the sixth CP1 architectural pattern. Distinguished from prior patterns by having many classes per single slot rather than many primitives per single class.

5. Falsifiability

Prediction 1994.1. SO_5^{24} should acquire additional current-domain instances at other SMBH targets (M87 SMBH at $\sim 10^9 M_{\text{sun}}$, TON 618 at $\sim 10^{10} M_{\text{sun}}$) once their CP1 stubs are drained. If not, the SO_5^{24} = SMBH DPM current lock may be specific to Sgr A* and not universal.

Prediction 1994.2. F_{TRZ}^6 rotation angular velocity should appear at other rotating astrophysical structures (BH accretion disks at intermediate mass, neutron star spin rates, galactic rotation curves) if the cross-domain interpretation is universal. Candidate test: any CP1 class with characteristic rotation $\omega \sim 10^{-6}$ rad/s.

Prediction 1994.3. $SO_5^9 = 1$ GHz frequency anchor should generalize beyond Sgr A* DPM/THz to other UHF-band astrophysical signals. Candidates: pulsar radio emission (~ 1.4 GHz L-band close to slot 9), CMB spectral shoulder, radio galaxy peaks.

Prediction 1994.4. N-class same-slot family patterns should surface at other well-populated ladder slots. Candidates: SO_5^3 (velocity + structural), SO_5^7 (timescale + mass + frequency + potential future). If additional slots reach 5+ class instances, the "N-class same-slot family" architecture is confirmed as a general CP1 pattern rather than a slot-21-specific anomaly.

6. Framework annotations

- **Backbone:** Sgr A SMBH SO_5^{24} first current-domain application (extends PAPER_1908 dimensionless-ratio use) + F_{TRZ}^6 SMBH rotation angular velocity cross-domain instance (extends PAPER_1985 $n=6$ rung to rotation domain) + $SO_5^9 = 1$ GHz first frequency-domain anchor (fills PAPER_1990 ladder gap between slots 7 and 10) + seven-class SO_5^{21} family (richest slot in PAPER_1955 ladder)
- **Method:** cross-domain applications of established primitive-arithmetic constants + gap-fill of pre-existing ladder + N-class same-slot family enumeration
- **Shells:** Sgr A SMBH DPM current-vortex + accretion disk THz-pipeline + reactive U_{g4i} shells + Compressed sector systems 18-24
- **CPCH:** CP1 Sgr A* SMBH sector (SgrAFreqDPM + SgrAFreqTHz + SgrAFreqUg4i R132 fills) + CP1 Compressed sector (7-class SO_5^{21} family cumulative through R132)
- **Spine:** PAPER_1908 SO_5^{24} seminal (dimensionless) + PAPER_1985 F_{TRZ}^6 seminal (magnetic field) + PAPER_1990 frequency-domain SO_5 -power ladder + PAPER_1955 SO_5^{21} slot + PAPER_1991 R129 same-round twin + PAPER_1993 R130 three-class family extension

- **Time frame:** quasi-static SMBH accretion resonance + cumulative same-slot family growth over R117-R132

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NOT REPLACEMENT. Offered as an alternative parameter-economical description ("NOT REPLACEMENT") to Standard Model + Lambda-CDM, with honest residuals reported alongside each closure.

References

- **PAPER_1908** — Q_{UQFF} SCm resonator quality factor + $Q^{-2} = \rho_{SCm} \times SO_5^{24}$ EXACT structural identity (SO_5^{24} seminal as dimensionless-ratio primitive)
- **PAPER_1985** — R117 dual discovery: Pillars F_{TRZ}^6 magnetic-field + NGC 2525 SMBH mass slot 7 (F_{TRZ}^6 seminal at $n=6$ rung, magnetic field domain)
- **PAPER_1990** — SO_5 -power frequency ladder cross-domain extension (slots 7 and 10 seminal, slot 9 gap filled by R132)
- **PAPER_1955** — SO_5 -power galactic structural ladder (SO_5^{21} slot originator)
- **PAPER_1991** — R129 base + revision addendum (SO_5^{21} same-round twin seminal + QUAD-lock architecture)
- **PAPER_1993** — R130 cross-rung $F_{TRZ} + 2\pi \cdot H_0$ Hubble + SO_5^{21} three-class family (predecessor to R132 seven-class extension)
- **PAPER_1919** — F_{TRZ} power ladder ($n=6$ rung status + cross-domain N-regime criteria)
- **PAPER_1868** — Solar physics (F_{TRZ}^6 -adjacent solar cycle instance)
- **PAPER_1945** — Magnetar $n_{lobes} \times F_{TRZ}$ universality (F_{TRZ}^6 magnetar instance)
- **PAPER_1976** — HUDF $I_0 + \tau_{inter} = SO_5^9$ (timescale-domain SO_5^9 instance for slot 9 timescale role)
- **PAPER_1952** — Galaxy-scale SO_5 -power timescale hierarchy (predicts $SO_5^9 = 1$ Gyr galaxy quenching timescale)
- **PAPER_1938** — $\omega_{SCm} = 1.25$ THz universal carrier family